DETAIL SPECIFICATION SHEET

# ADAPTERS, COAXIAL TO WAVEGUIDE
# (SERIES N TO WAVEGUIDE - CLASS 1)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the adapters described herein shall consist of this document and the latest issue of MIL-DTL-22641.

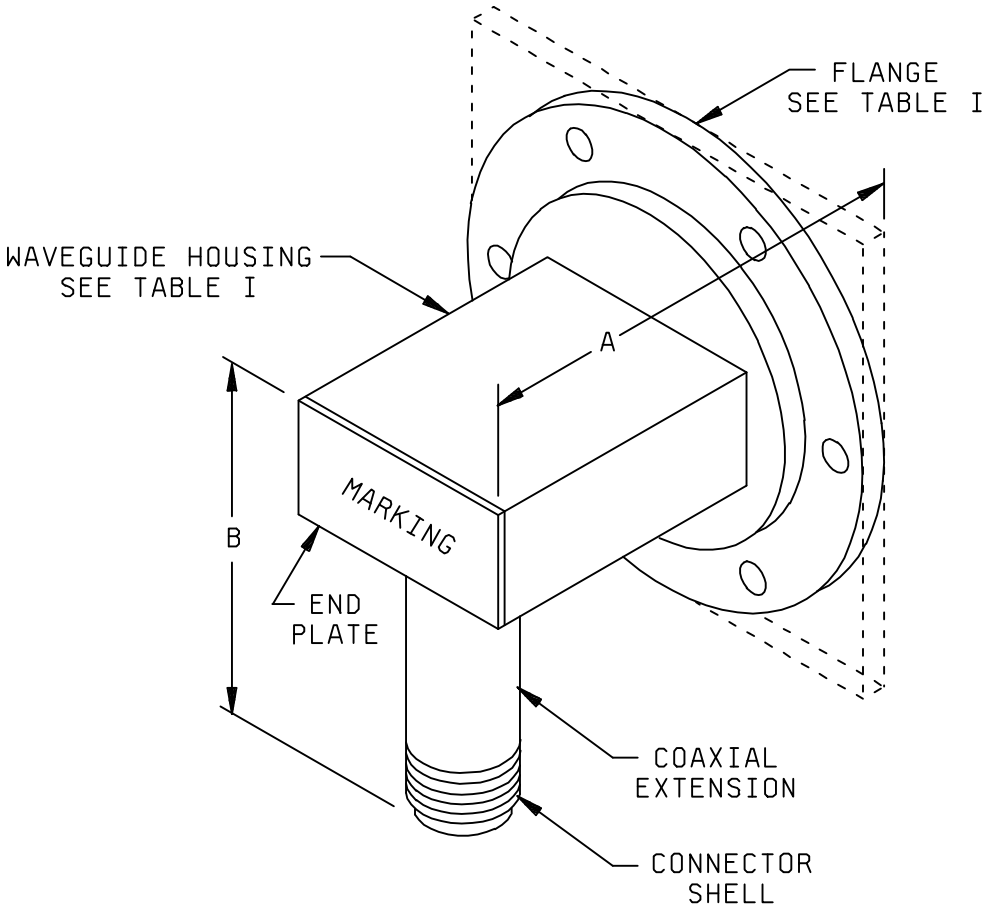


FIGURE 1. Configuration for PIN M22641/16-01 through M22641/16-22.


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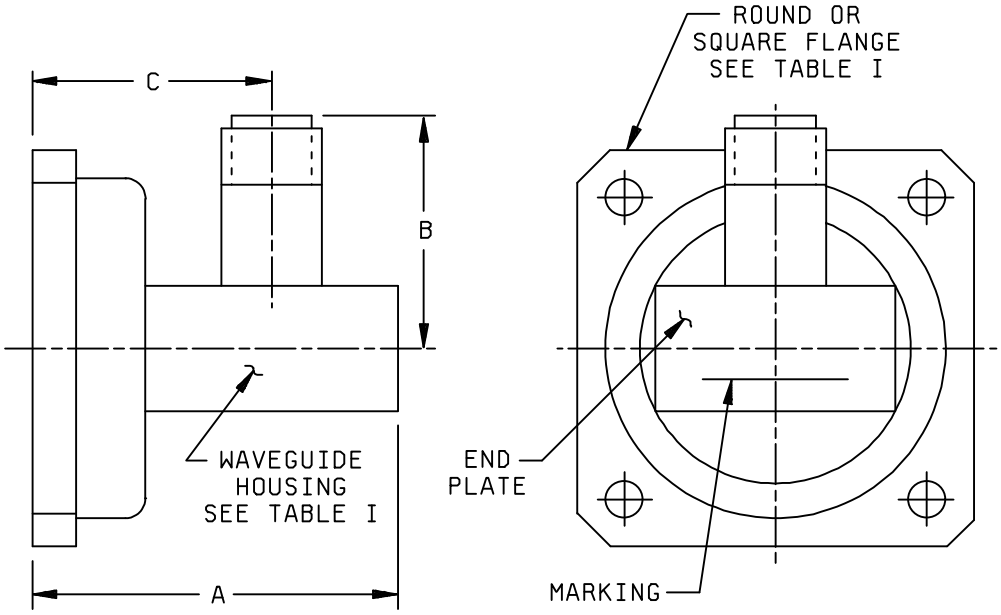


FIGURE 2. Configuration for PIN M22641/16-23 through M22641/16-30.

REQUIREMENTS:

Design, construction, and physical dimensions: See figures 1 and 2 and table I.

Frequency range: See table I.

VSWR: See table I. VSWR over 50 percent frequency range shall be VSWR measured over any continuous frequency band of width equal to 50 percent of the entire frequency range, chosen at the discretion of the tester.

RF insertion loss: See table I.

Radio frequency connector: The coaxial connector shall be compatible and mate with series N plug in accordance with MIL-PRF-39012/1.

Flange, waveguide: See table I.

Housing, waveguide: See table I.



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TABLE I. Adapters, coaxial to waveguide (class I - series N to waveguide).

| Dash number | Figure | Frequency range (GHz) | VSWR over entire frequency range 5/ 6/ | VSWR over 50 percent frequency range 5/ 6/ | RF insertion loss (dB) max 6/ | Waveguide housing 7/ | Flange 3/ | Dimensions 1/ 2/ | | | Waveguide and flange material |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | A (max) | B (max) | C (max) | |
| 01 | 1 | 7.05 - 10.0 | 1.25:1 | 1.20:1 | .10 | M85/1-069<br>M85/1-073<br>RG-51/U | M3922/53-002<br>UG-51/U<br>Cover 3/ | 2.197<br>(55.80) | 2.250<br>(57.15) | --- | Copper alloy |
| 02 | 1 | 5.85 - 8.2 | 1.25:1 | 1.20:1 | .10 | M85/1-063<br>M85/1-067<br>RG-50/U | M3922/55-001<br>UG-344/U<br>Cover 3/ | 2.500<br>(63.50) | 2.375<br>(60.33) | --- | Copper alloy |
| 03 | 1 | 3.95 - 5.85 | 1.25:1 | 1.20:1 | .10 | M85/1-051<br>M85/1-055<br>RG-49/U | M3922/57-002<br>UG-149A/U<br>Cover 3/ | 3.687<br>(93.65) | 2.437<br>(61.90) | --- | Copper alloy |
| 04 | 1 | 2.6 - 3.95 | 1.25:1 | 1.20:1 | .10 | M85/1-039<br>M85/1-043<br>RG-48/U | M3922/56-001<br>UG-53/U<br>Cover 3/ | 4.062<br>(103.17) | 4.250<br>(107.95) | --- | Copper alloy |
| 05 | 1 | 8.2 - 12.4 | 1.25:1 | 1.20:1 | .20 | M85/1-079<br>M85/1-075<br>RG-52/U | M3922/53-001<br>UG-39/U<br>Cover 3/ | 2.000<br>(50.80) | 2.218<br>(56.34) | --- | Copper alloy |
| 06 | 1 | 1.7 - 2.6 | 1.25:1 | 1.20:1 | .05 | M85/1-027<br>M85/1-031<br>RG-104/U | M3922/58-009<br>UG-435B/U<br>Contact 3/ | 4.125<br>(104.78) | 5.000<br>(127.00) | --- | Copper alloy |
| 07 | 1 | 1.12 - 1.7 | 1.25:1 | 1.20:1 | .05 | M85/1-015<br>M85/1-017<br>RG-69/U | M3922/58-007<br>UG-417B/U<br>Contact 3/ | 5.250<br>(133.35) | 7.250<br>(184.15) | --- | Copper alloy |
| 08 | 1 | 4.9 - 7.O5 | 1.25:1 | 1.20:1 | .10 | M85/1-057<br>M85/1-061<br>RG-343/U | M3922/52-037<br>UG-1354/U<br>Contact 3/ | 2.000<br>(50.80) | 2.000<br>(50.80) | --- | Copper alloy |
| 09 | 1 | 7.05 - 10.0 | 1.25:1 | 1.20:1 | .10 | M85/1-071<br>RG-68/U | M3922/53-004<br>UG-138/U<br>Cover 3/ | 2.197<br>(55.80) | 2.250<br>(57.15) | --- | Aluminum alloy |

See footnotes at end of table.



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TABLE I. Adapters, coaxial to waveguide (class I - series N to waveguide) - Continued.

| Dash number | Figure | Frequency range (GHz) | VSWR over entire frequency range 5/ 6/ | VSWR over 50 percent frequency range 5/ 6/ | RF insertion loss (dB) max 6/ | Waveguide housing 7/ | Flange 3/ | Dimensions 1/ 2/ | | | Waveguide and flange material |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | A (max) | B (max) | C (max) | |
| 10 | 1 | 5.85 -8.2 | 1.25:1 | 1.20:1 | .10 | M85/1-065<br>RG-106/U | M3922/55-002<br>UG-441/U<br>Cover 3/ | 2.500<br>(63.50) | 2.375<br>(60.33) | --- | Aluminum alloy |
| 11 | 1 | 3.95 - 5.85 | 1.25:1 | 1.20:1 | .10 | M85/1-053<br>RG-95/U | M3922/57-001<br>UG-407/U<br>Cover 3/ | 3.687<br>(93.65) | 2.437<br>(61.90) | --- | Aluminum alloy |
| 12 | 1 | 2.6 - 3.95 | 1.25:1 | 1.20:1 | .10 | M85/1-041<br>RG-75/U | M3922/56-002<br>UG-584/U<br>Cover 3/ | 4.062<br>(103.17) | 4.250<br>(107.95) | --- | Aluminum alloy |
| 13 | 1 | 8.2 - 12.4 | 1.25:1 | 1.20:1 | .20 | M85/1-077<br>RG-67/U | M3922/53-003<br>UG-135/U<br>Cover 3/ | 2.000<br>(50.80) | 2.218<br>(56.34) | --- | Aluminum alloy |
| 14 | 1 | 1.7 - 2.6 | 1.25:1 | 1.20:1 | .10 | M85/1-029<br>RG-105/U | M3922/58-010<br>UG-437B/U<br>Contact 3/ | 4.125<br>(104.78) | 5.000<br>(127.00) | --- | Aluminum alloy |
| 15 | 1 | 1.12 - 1.7 | 1.25:1 | 1.20:1 | .05 | M85/1-018<br>RG-103/U | M3922/58-008<br>UG-418B/U<br>Contact 3/ | 5.250<br>(133.35) | 7.250<br>(184.15) | --- | Aluminum alloy |
| 16 | 1 | 4.9 - 7.05 | 1.25:1 | 1.20:1 | .10 | M85/1-059<br>RG-344/U | M3922/52-038<br>UG-1355/U<br>Contact 3/ | 2.000<br>(50.80) | 2.000<br>(50.80) | --- | Aluminum alloy |
| 17 | 1 | 7.0 - 11.0 | 1.25:1 | 1.20:1 | .20 | M85/1-155<br>RG-320/U | M3922/70-001<br>UG-1493/U<br>Cover 3/ | 1.515<br>(38.48) | 1.755<br>(44.58) | --- | Copper alloy |
| 18 | 1 | 4.9 - 7.05 | 1.25:1 | 1.20:1 | .10 | M85/1-059<br>M85/1-060<br>RG-344/U | M3922/52-038<br>UG-1355/U<br>Contact 3/ | 3.500<br>(88.90) | 2.188<br>(55.58) | --- | Aluminum alloy |

See footnotes at end of table.



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TABLE I. Adapters, coaxial to waveguide (class I - series N to waveguide) - Continued.

| Dash number | Figure | Frequency range (GHz) | VSWR over entire frequency range 5/ 6/ | VSWR over 50 percent frequency range 5/ 6/ | RF insertion loss (dB) max 6/ | Waveguide housing 7/ | Flange 3/ | Dimensions 1/ 2/ | | | Waveguide and flange material |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | A (max) | B (max) | C (max) | |
| 19 | 1 | 4.9 - 7.05 | 1.25:1 | 1.20:1 | .10 | M85/1-061 RG-343/U | M3922/52-037 UG-1354/U Contact 3/ | 3.500 (88.90) | 2.188 (55.58) | --- | Copper alloy |
| 20 | 1 | 2.25 - 5.5 | 1.30:1 | 1.20:1 | .10 | M85/4-001 RG-109/U | M3922/75-01 UG-509/U Cover | 6.000 (152.40) | 4.000 (101.60) | --- | Copper alloy |
| 21 | 1 | 5.0 - 10.0 | 1.40:1 | 1.30:1 | .10 | M85/4-003 RG-110/U | M3922/75-03 UG-511/U Cover | 3.00 (76.2) | 2.75 (69.9) | --- | Copper alloy |
| 22 4/ | 1 | 7.0 - 11.0 | 1.10:1 | 1.10:1 | .20 | M85/1-155 RG-320/U | M3922/70-001 UG-1493/U 3/ Cover | 1.40 (35.6) | 1.10 (27.9) | --- | Copper alloy |
| 23 | 2 | 3.95 - 5.85 | 1.065:1 | 1.03:1 | .10 | M85/1-055 RG-49/U | M3922/52-013 UG-1728/U Contact 3/ | 2.00 (50.8) | 1.96 (49.8) | 1.50 (38.1) | Copper alloy |
| 24 | 2 | 3.95 - 5.85 | 1.065:1 | 1.03:1 | .10 | M85/1-053 RG-95/U | M3922/52-014 UG-1729/U Contact 3/ | 2.00 (50.80) | 1.96 (49.8) | 1.50 (38.1) | Aluminum alloy |
| 25 | 2 | 7.05 - 10.00 | 1.065:1 | 1.03:1 | .10 | M85/1-073 RG-51/U | M3922/52-019 UG-1734/U Contact 3/ | 1.44 (36.6) | 1.07 (27.2) | .88 (22.4) | Copper alloy |
| 26 | 2 | 7.05 - 10.00 | 1.065:1 | 1.03:1 | .10 | M85/1-071 RG-68/U | M3922/52-020 UG-1735/U Contact 3/ | 1.44 (36.6) | 1.07 (27.2) | .88 (22.4) | Aluminum alloy |
| 27 8/ | 2 | 2.60 - 3.95 | 1.04:1 | --- | .05 | M85/4-017 | M3922/75-06 | 2.50 (63.5) | 1.29 (32.8) | 1.53 (38.9) | Aluminum alloy |

See footnotes at end of table.



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TABLE I. Adapters, coaxial to waveguide (class I - series N to waveguide) - Continued.

| Dash number | Figure | Frequency range (GHz) | VSWR over entire frequency range 5/ 6/ | VSWR over 50 percent frequency range 5/ 6/ | RF insertion loss (dB) max 6/ | Waveguide housing 7/ | Flange 3/ | Dimensions 1/ 2/ | | | Waveguide and flange material |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | A (max) | B (max) | C (max) | |
| 28 8/ | 2 | 2.60 - 5.85 | 1.25:1 max | --- | .05 | M85/4-007 | M3922/75-02 | 2.50 (63.5) | 1.47 (37.3) | 1.60 (40.6) | Aluminum alloy |
| 29 8/ | 2 | 5.85 - 8.20 | 1.04:1 | --- | .05 | M85/4-008 | M3922/75-04 | 1.50 (38.1) | 1.17 (29.7) | 1.13 (28.7) | Aluminum alloy |
| 30 8/ | 2 | 5.85 - 12.4 | 1.25:1 max | --- | .05 | M85/4-008 | M3922/75-04 | 1.50 (38.1) | 1.17 (29.7) | 1.13 (28.7) | Aluminum alloy |

1/ Dimensions are in inches.
2/ Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3/ Flange in accordance with MIL-F-3922.
4/ Connector shall be female type N (corrosion-resistant steel). Shock, vibration, and salt spray requirements are not required for this Part or Identifying Number (PIN).
5/ A larger attenuator pad may be used for VSWR test setup (see figure 1 of MIL-DTL-22641) when measuring VSWR values less than 1.25:1.
6/ Swept frequency measuring technique is an acceptable alternate test method for determining VSWR and RF insertion loss value.
7/ Waveguide in accordance with MIL-DTL-85.
8/ Reduced height waveguides and flanges.



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Marking: See figures 1 and 2.

PIN: M22641/16- (dash number from table I).

TABLE II. Cross-reference of AN nomenclature to PIN.

| AN nomenclature | PIN M22641/16- ( ) | AN nomenclature | PIN M22641/16- ( ) |
|---|---|---|---|
| UG-397/U | 01 | UG-1653/U | 09 |
| UG-398/U | 02 | UG-1654/U | 10 |
| UG-399/U | 03 | UG-1655/U | 11 |
| UG-400/U | 04 | UG-1656/U | 12 |
| UG-446A/U | 05 | UG-1657/U | 13 |
| UG-952/U | 06 | UG-1658/U | 14 |
| UG-953/U | 07 | UG-1659/U | 15 |
| UG-1004/U | 20 | UG-1660/U | 16 |
| UG-1005/U | 21 | UG-1661/U | 08 |
| UG-1538/U | 17 | UG-1679/U | 19 |
| UG-1646/U | 18 | | |

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, GL, MI
Navy - AS, OS
Air Force - 99

Preparing activity:
DLA - CC

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